Love Under Pressure: Caring for Aging Parents and the Risk of Caregivers' Marital Dissolution *

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November 3, 2026

Abstract

Caring for aging parents has been known to adversely impact family caregivers' labor supply and health. These effects potentially spill over into marital relationships. Yet the relationship between eldercare and caregivers' marital dissolution remains understudied. Using longitudinal data from the Health and Retirement Study (HRS), I combine propensity score matching with Cox proportional hazards model to estimate this relationship. I find that caring for aging parents is associated with an approximately 51 percent higher hazard of separation or divorce. Mechanism analysis provides suggestive evidence that the association may be related to poorer mental health, weaker labor market attachment, and lower marital satisfaction among caregivers. The estimated effect seems to be stronger among male caregivers, compared to female caregivers. This study provides the first quantitative evidence of the relationship between parental caregiving and adult caregivers' marital dissolution, highlighting family stability as an overlooked cost of policies that shift long-term care from formal providers to family members.

Keywords: Family Caregivers, Divorce, Informal Care, Marital Quality **JEL Codes:** I12, J12, J13

* The author thanks Dr. Shooshan Danagoulian and Dr. Yulya Truskynovsky for their mentorship, as well as participants in the Wayne State Dissertation Seminar for valuable feedback.

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1 Introduction

Divorce in the United States is increasingly concentrated in the same stage of life when adults are most likely to assume caregiving responsibilities for aging parents. Divorces involving adults aged 50 and older accounted for 36 percent of all divorces in the United States. Adults aged 65 and older are the only age group for whom the divorce rate continues to rise (Brown and Lin, 2022). At the same time, this period of the life also overlaps with the years when adults are most likely to confront the caregiving needs of their aging parents. This demand has intensified as life expectancy has increased and as more adults find themselves simultaneously managing careers, children, and parental care in the so-called "sandwich generation". Despite the temporal overlap between these two demographic trends, the literature on rising gray divorce has largely overlooked parental caregiving responsibilities as a potential contributor to marital instability. More broadly, little is known about the predictors of marital instability in middle and later life (Brown and Lin, 2012).

Caregiving for an aging parent is theoretically capable of straining caregivers' marriage through several channels. The first channel is time and emotional strain, as caregiving consumes hours and energy that would otherwise sustain the marital relationship. The second channel is financial strain, arising from reduced labor supply, foregone earnings, or out-of-pocket medical expenses associated with parental care. The third channel is interpersonal conflict, as spouses disagree over the allocation of caregiving responsibilities or the caregiver's divided attention.

Existing research on caregiving and marital dissolution falls into three separate strands, none of which directly studies whether caregiving translates into divorce. The first strand documents the psychological and relational burden of parental caregiving without examining the behavioral transition into divorce itself. For example, Amirkhanyan and Wolf (2006) find that having a parent in need of care increases depressive symptoms, Bookwala (2009) shows that experienced caregivers report lower marital happiness, greater marital

role inequity, and more hostility than caregivers who have recently assumed the role, and (Choi and Marks, 2006) finds that caregiving for spouses and parents is associated with greater marital disagreement. As Schulz et al. (2016) explicitly acknowledges, family relationships affected by caregiving demands have "received relatively little attention in the caregiving literature," and even studies that do address relationship quality stop short of tracking actual divorce over time.

The second strand establishes an identification strategy: population-based propensity matching combined with Cox proportional hazards models, as used by Roth et al. (2013), shows that caregivers experience different mortality hazards than comparable non-caregivers, but this approach has been applied almost exclusively to survival/mortality outcomes, not to marital outcomes.

The third strand meticulously explains demographic drivers of gray divorce but treats caregiving responsibilities as outside its explanatory scope (Brown and Lin, 2012, 2022). Taken together, these three strands leave open the same central question: whether caregiving-induced marital strain actually translates into caregivers' divorce.

A central empirical challenge is that caregiving is not randomly assigned. Individuals who become caregivers may differ systematically from non-caregivers in characteristics that also affect marital stability. Differences in employment, economic resources, family structure, and other observable characteristics may therefore confound comparisons between caregivers and non-caregivers. In addition, unobserved factors may influence both the likelihood of assuming caregiving responsibilities and the risk of divorce. Without accounting for these sources of selection, estimates of the relationship between caregiving and divorce may be biased.

This paper addresses this gap by providing new evidence on the relationship between caring for aging parents and marital dissolution. Using longitudinal data from the Health and Retirement Study (HRS), I examine whether parental caregiving is associated with the risk and timing of marital dissolution. Methodologically, I combine propensity score

matching (PSM) with Cox proportional hazards models to account for observable differences between caregivers and non-caregivers and to examine the association between caregiving status and the hazard of marital dissolution over time.

This study supports a straightforward conclusion: caring for aging parents is associated with a statistically significant increase in the hazard of divorce. The estimated effect is substantively meaningful: caregivers face a hazard of divorce approximately 51 percent higher than comparable non-caregivers. This finding is robust to alternative matching methods, alternative survival model specification, and sensitivity analyses for unobserved confounding.

Marital quality, mental health, and employment attachment are explored as potential mechanisms driving the main finding on divorce. I find a statistically significant negative impact of eldercare on caregivers' mental health. There is suggestive evidence of worse marital ratings, measured by whether caregivers report feeling close to, understood by, open with, or let down by their spouse. Caregivers are also less likely to work full-time and more likely to leave the labor force. Heterogeneity analysis suggests gender and education status also shape this relationship. Specifically, the estimated increase in divorce hazard is larger among male caregivers and individuals with higher educational attainment.

This paper contributes to several branches of literature. First, it contributes to the literature on informal caregiving burden by identifying marital dissolution as an important potential consequence of caring for aging parents. Prior work has treated divorce either as a marital-quality proxy or as an anecdotally noted extreme consequence, rather than as a directly observed outcome (Schulz et al., 2016). This study instead provides the first quantitative evidence on this relationship, moving it from a qualitatively acknowledged possibility to an empirically tested outcome.

Second, in terms of empirical method, this study combines matching method with a Cox proportional hazards framework to estimate the effect of eldercare on divorce.

PSM balances observable differences in the propensity to become a caregiver, while the Cox framework quantifies the resulting effect on divorce hazard over time. This same PSM–Cox combination has previously demonstrated its value in studying caregiving effects on mortality, but it has not yet been extended to marital outcomes in measuring association to eldercare.

Third, by empirically linking caregiving to divorce, this paper also bridges two literatures that have so far developed in isolation. The first is the caregiving stress-process literature, which documents the psychological toll of parent care (Amirkhanyan and Wolf, 2006; Bookwala, 2009). The second is the demographic literature on gray divorce, which has documented a doubling in divorce rates among adults aged 50 and older between 1990 and 2010 (Brown and Lin, 2012, 2022). This literature has not incorporated caregiving responsibilities as an explanatory mechanism, an omission worth noting given the substantial overlap in the life-course timing of both phenomena.

The remainder of this paper proceeds as follows. Section 2 provides background on eldercare and divorce. Section 3 describes data, sample selection and key variables. Section 4 shows comparable group construction and Cox PH model. Section 5 presents results. Finally, section 6 discusses the implications of the findings and concludes.

2 Background

2.1 Caregiving for aging parents

Family caregiving becomes reality of aging population and affordable long-term care shortage. In the United States, nearly one in four individuals serves as a family caregiver, with the majority providing care to adults and doing so without financial compensation (National Family Caregiver Support Program, 2026). Informal caregiving represents a substantial economic contribution. It is valued at approximately $1.01 trillion annually, with caregivers providing an estimated 49.5 billion hours of care-equivalent to about 17%

of the full-time workforce.

A large literature documents adverse consequences of caregiving along two dimensions: labor market outcomes and psychological well-being. On the labor market side, caregivers are more likely to reduce their labor supply, retire, or experience lower earnings, with these effects being particularly pronounced among individuals who provide care more frequently or intensively (Van Houtven et al., 2013; Jacobs et al., 2019; Truskinovsky and Maestas, n.d.; Ciccarelli and Van Soest, 2018). These effects may also persist beyond the caregiving episode, with reductions in full-time employment and long-run earnings observed several years after caregiving has ended (Schmitz and Westphal, 2017; Heger and Korfhage, 2020), and are not uniform across caregivers: women tend to experience larger labor-market consequences than men, including greater reductions in employment and working hours (Van Houtven et al., 2013; Smith et al., 2020).

On the psychological side, caregiving has been linked to higher levels of depression, anxiety, stress, and poorer quality of life, particularly when caregiving is intensive or prolonged. Providing care to parents is associated with increases in depressive symptoms and declines in self-rated health among married men and women, with particularly persistent effects on depression among married women (Coe and Van Houtven, 2009), and more intensive caregiving is associated with greater psychological distress that may persist even after caregiving ends (Hirst, 2005).

Evidence on labor and health outcomes suggests caring for an aging parent requires time and resources that could otherwise be devoted to caregivers' marital relationship. Reduced employment and earnings may increase financial pressure within the household. Caregiving-related stress may affect how spouses interact with one another. Unlike spousal caregiving, in which caregiving occurs within the marital relationship itself, parental caregiving involves an obligation to a family member outside the marital dyad, creating competing demands between responsibilities to aging parents and responsibilities within the caregiver's own household.

Family relationships affected by caregiving demands for aging parents have "received relatively little attention in the caregiving literature" (Schulz et al., 2016). To date, no study has examined separation or divorce as an outcome associated with parental caregiving in the US. Outcomes were explored includes: marital happiness, marital role inequity and marital disagreement (Bookwala, 2009).

2.2 Marital Dissolution

Divorce represents an important form of marital instability with substantial consequences for individuals, households, and families. A useful theoretical framework for understanding its determinants is Becker's theory of marriage, which conceptualizes marital stability in terms of marital surplus: individuals remain married when the benefits of marriage exceed the value of available alternatives, and shocks that reduce household productivity, increase household costs, or alter the distribution of resources within the household can reduce this surplus and increase the likelihood of marital dissolution (Becker, 1974). From this perspective, caregiving for an aging parent can be viewed as a household shock that operates through the same channels documented above - reduced time, financial resources, and psychological well-being - each of which lowers the gains from remaining married.

Consistent with this framework, prior research links each of these channels to divorce independently. Financial hardship, household debt, and job loss increase marital strain and the likelihood of separation or divorce (Butler et al., n.d.; Bridges and Disney, 2016; Guo et al., 2024), which is particularly relevant to parental caregiving given its documented effects on labor supply and earnings. Mental health and psychological distress are similarly important: mental distress in either spouse is associated with a higher risk of subsequent divorce (Idstad et al., 2015), and mood and anxiety disorders have been linked to a greater likelihood of marital dissolution over time (Mojtabai et al., 2017). Because caregiving increases depression, stress, and other forms of psychological distress,

these findings provide a further pathway linking parental caregiving to divorce, though the relationship between mental health and divorce may also be bidirectional, underscoring the importance of examining the temporal ordering between caregiving and marital dissolution.

Despite this theoretical and empirical groundwork, relatively little is known about whether caring for aging parents is directly associated with the timing of marital dissolution among married adults in the United States. Existing research has primarily examined caregivers' health, labor-market outcomes, and economic well-being in isolation, while studies of marital consequences have more often focused on relationship quality, disagreement, or union dissolution in other populations, leaving the direct link between parental caregiving and divorce, and its longitudinal timing, largely unexamined. This study addresses this gap using longitudinal data from the Health and Retirement Study, following respondents from marriage through subsequent separation or divorce while accounting for demographic, socioeconomic, health, and family characteristics, and examining marital quality and mental health as potential mechanisms linking caregiving to divorce.

2.3 Section into caregiving

Caregivers and non-caregivers differ across a range of demographic and socioeconomic characteristics. Table A1 presents summary statistics by caregiving status. On average, caregivers are younger than non-caregivers (62.2 versus 67.4 years). They are also more likely to be female (63.7% compared to 50.5%) and to have higher levels of educational attainment. In terms of labor market outcomes, caregivers are more likely to be employed full-time (34.1% compared to 25.0%). Household composition differs slightly, with caregivers having marginally fewer children in the household (3.29 versus 3.45).

The observed differences between caregivers and non-caregivers may lead to biased estimates if these characteristics are correlated with both caregiving status and divorce

outcomes. For example, female are more likely to initiate divorce (Kalmijn and Poortman, 2006) and more likely to take the caregiving responsibility for mother (Leopold et al., 2014). Divorce decision may be influenced by their own education attainment and that of their parent (Lyngstad, 2004). Becker et al. (1977) developed a theory framework explaining the impact of multiple factors including the number of children on marital stabilty. According to their framework, the number of children in the household impacts the divorce decision. Specifically, woman are less likely to divorce if they have higher number of children under age five. The effect fades when the number of older kids increases. The ability to work full time impacts household financial constrain, which is among factors leading to marital dissolution (Nunley and Seals, 2010). These studies illustrate the effect of socio-demographics on divorce. As a result, naïve comparisons of the probability of divorce between caregivers and non-caregivers may conflate the effect of caregiving with pre-existing differences across individuals, leading to selection bias.

3 Data

3.1 Data Source

This study uses publicly available longitudinal data from the Health and Retirement Study (HRS) covering the period from 1992 to 2020. The HRS is a nationally representative panel survey of U.S. adults that is administered biennially. The analytic dataset is constructed by merging three HRS data sources: the RAND HRS Longitudinal File (2020), the RAND HRS Family Data (2020), and the HRS Core Leave-Behind Questionnaire files from 2004 through 2020.

The RAND HRS Longitudinal File (2020) provides a comprehensive set of individual- and household-level characteristics. These include labor force status, mental health, marital status, age, gender, education, net worth, place of birth, race and parental characteristics such as parental education. The RAND HRS Family Data (2020) supplements these

measures with information on family structure and intergenerational relationships. In particular, it provides measures of the number of children in the family and respondents' informal caregiving activities, including whether they provided care to their parents and the number of hours spent providing such care since the previous interview. Finally, the HRS Core Leave-Behind Questionnaire provides measures of self-reported marital quality, capturing several dimensions of respondents' relationships with their spouses, including understanding, reliance, openness, closeness, and feeling let down.

These data sources are merged using the HRS respondent identifiers to construct a longitudinal panel in which respondents can be followed across biennial interview waves. The resulting panel allows the analysis to examine changes in caregiving status and marital outcomes over time while accounting for respondents' demographic, socioeconomic, health, family, and relationship characteristics.

3.2 Sample Selection

The analytic sample is constructed using two primary inclusion criteria. First, respondents must be non-caregivers at the first wave in which they enter the analytic sample. Establishing a common non-caregiving baseline allows the analysis to compare respondents who subsequently become caregivers with those who remain non-caregivers from the same initial state. Second, respondents must be married at their baseline wave. This restriction is necessary because the primary outcome is the transition from marriage to separation or divorce, and the analysis of marital quality requires respondents to be in a marital relationship at baseline.

The resulting sample consists of 26,911 individuals and 191,110 person-wave observations. Among these respondents, 4,614 individuals provided eldercare at some point during the observation period, whereas 22,297 individuals never reported providing eldercare during the observed period. Summary statistics for the analytic sample are presented in Table A1.

3.3 Key Variables

3.3.1 Divorce

The primary outcome is the transition out of marriage through separation or divorce. I construct a binary indicator that equals one when a respondent is reported as separated or divorced at a given interview wave and zero when the respondent remains married or partnered. The outcome therefore captures the occurrence of marital dissolution during the observation period and serves as the event indicator in the subsequent survival analysis.

3.3.2 Caregiving for Aging Parents

The primary explanatory variable is caregiving for aging parents. Caregiving is defined based on the amount of time respondents report spending helping their parents with errands or basic personal needs, including dressing, eating, and bathing. A respondent is classified as a caregiver if they report providing any positive number of hours of care to a parent since the previous HRS interview. Respondents who did not report a positive caregiving hours are classified as non-caregivers for that observation period. This definition captures caregiving status at baseline rather than categorizing caregiving by intensity based on reported hours of care, due to concerns about the accuracy of self-reported hours.

3.3.3 Marital Quality

Marital quality is measured using respondents' self-reported assessments of several dimensions of their relationship with their spouse. Specifically, respondents report the extent to which their spouse: (1) understands the way they feel, (2) can be relied upon in serious situations, (3) is someone with whom they can openly discuss worries, (4) is emotionally close to them, and (5) lets them down. For each item, responses are coded on a

four-point scale: 1 (a lot), 2 (some), 3 (a little), and 4 (not at all).

3.3.4 Mental Health Score

To examine potential mechanisms linking caregiving to relationship outcomes, I also consider respondents' mental health, measured using the Center for Epidemiologic Studies Depression (CES-D) scale. The CES-D score (RwCESD) is constructed from six indicators of negative affect and two indicators of positive affect. The negative indicators capture whether respondents experienced depressive symptoms, including feeling depressed, feeling that everything was an effort, having restless sleep, feeling lonely, feeling sad, and feeling unable to get going, most or all of the time. The two positive indicators measure whether respondents felt happy and enjoyed life during the same period and are reverse-coded when constructing the index. The resulting CES-D measure ranges from 0 to 8, with higher values indicating a greater presence of depressive symptoms.

4 Research Design

4.1 Comparable non-caregiver group

To mitigate concerns about selection bias into caregiving for parents, I implement PSM method to construct a comparable group of non-caregivers. The matching is performed on a comprehensive set of socio-demographic covariates, including individual characteristics (age, gender, educational attainment, labor market status, race, and birthplace), parental characteristics (education), and household characteristics (number of children in the household and log(net worth)) at baseline wave. The goal of PSM method is to create a comparable group which have similar propensity score, helping to interpret the effect on the hazard of divorce through caregiving rather than difference in observable characteristics. Table 1 provides summary statistics for all person-wave observations in the matched sample, including waves observed after a divorce event.

I evaluate the performance of the propensity score matching (PSM) procedure using several diagnostic approaches. First, I compare the absolute mean differences in covariates between caregivers and non-caregivers before and after matching in Appendix figure A1. Second, I examine the distribution of propensity scores by caregiving status presented in Appendix figure A2. Third, I assess covariate balance using standardized bias measures and t-tests shown in Appendix table A2.

The above results indicate substantial improvements in balance following matching. In particular, the absolute mean differences across groups decrease for the majority of covariates. The distribution of propensity scores for the matched non-caregiver group closely mirrors that of the caregiver group, suggesting improved overlap. Moreover, the covariate balance tests show that across all thirteen covariates, the standardized percentage bias remains below 5%, ranging from 0.5% for depressive symptoms to 4.4% for the White indicator, well within the conventional threshold used to assess adequate covariate balance. Consistent with this, t-tests reveal no statistically significant mean differences between the treated and control groups for any covariate, with p-values ranging from 0.193 to 0.888. These evidence suggest that the PSM procedure constructs a more comparable control group relative to the unmatched sample.

4.2 Cox Proportional Hazard Model

To estimate the effect of caregiving for aging parents on caregivers' divorce rate, I employ the Cox PH model with time-varying covariates. The hazard function is specified as follows:

$$h_i(t \mid X_i(t)) = h_0(t) \exp\left(\beta_1 \, Caregiving_i(t) + \mathbf{X}_i(t)'\boldsymbol{\gamma}\right) \tag{1}$$

where $h_i(t \mid X_i(t))$ denotes the hazard of divorce for individual i at time t, conditional on the covariates $X_i(t)$, and $h_0(t)$ is the unspecified baseline hazard function, common to

all individuals. $Caregiving_i(t)$ is an indicator variable equal to one if individual i provides care to a parent as of wave t, and zero otherwise, allowing an individual's caregiving status to switch on or off across waves. $\mathbf{X}_i(t)$ is a vector of individual, household, and parental characteristics, as listed in summary statistic Table 1, which may likewise vary across waves, and γ is the corresponding vector of coefficients.

The coefficient of interest, β_1, represents the log hazard ratio associated with caregiving. A positive value of β_1 indicates that, at any given wave, being in a caregiving spell is associated with a higher hazard of divorce relative to not caregiving, holding the covariates in $\mathbf{X}_i(t)$ constant. The hazard ratio, $HR = \exp(\beta_1)$, provides a more directly interpretable measure of this association: an HR greater than one indicates that caregivers face a higher instantaneous risk of divorce than non-caregivers at any given point in analysis time, while an HR less than one would indicate a lower risk. Standard errors are clustered at the individual level to account for repeated observations of the same individual across survey waves.

The key assumption underlying the Cox model is the proportional hazards assumption. This assumption states that the hazard ratio associated with caregiving remains constant over analysis time. This assumption is what allows the Cox model to leave the baseline hazard $h_0(t)$ unspecified. As a result, the model avoids imposing a parametric distributional assumption on the timing of divorce.

I assess this assumption using a Schoenfeld residual test. For each individual who experiences divorce, this test computes the difference between the individual's observed value of the caregiving covariate and a weighted average of that covariate among all individuals still at risk of divorce at that same point in time. If the proportional hazards assumption holds, these residuals should show no systematic pattern over time; they should fluctuate randomly around zero regardless of when in the follow-up period the divorce occurs. If, instead, the effect of caregiving on divorce risk strengthens or weakens as analysis time progresses, the residuals will trend upward or downward over time

rather than remaining flat.

Figure 1 presents the scaled Schoenfeld residuals for the caregiving indicator, plotted against the rank of analysis time. Each point represents the residual for an individual who experienced divorce, and the solid line represents a smoothed estimate of these residuals over time. The smoothed line remains close to a value of approximately 0.4 throughout the entire range of analysis time, from the earliest to the latest ranks. There is no visible upward or downward trend in the smoothed estimate as analysis time progresses. This flat pattern indicates that the association between caregiving and the hazard of divorce does not systematically strengthen or weaken over the follow-up period. This visual evidence is supported by the formal Schoenfeld residuals test result presented in Appendix table A3. The correlation between the scaled Schoenfeld residuals and the rank of analysis time is small and statistically insignificant ($\rho = -0.029$, $\chi^2 = 0.26$, $p = 0.608$), providing no evidence that the effect of caregiving varies systematically over time. The global test across all covariates is also statistically insignificant ($\chi^2 = 16.22$, $p = 0.367$), indicating no evidence of a violation of the proportional-hazards assumption for the model as a whole. Collectively, this pattern is consistent with the proportional hazards assumption: the effect of caregiving on the hazard of divorce appears stable across analysis time.

5 Results

5.1 The effect of eldercare on caregivers' divorce

This section presents the effect of caregiving for aging parents on caregivers' divorce. I begin with a descriptive analysis using Kaplan-Meier survival curves, shown in Figure 3. This graph compares the estimated probability of remaining in marriage at each point in analysis time between caregivers and non-caregivers. In this figure, the horizontal axis measures analysis time in waves, and the vertical axis reports the corresponding survival probability. The two groups exhibit broadly similar survival patterns during the early

waves, indicating no immediate divergence in divorce risk between caregivers and non-caregivers. Beginning around wave 9, however, the survival curve for caregivers starts to decline more rapidly, and the gap between the two groups widens over time. These descriptive Kaplan-Meier estimates suggest that caregivers experience divorce earlier and more frequently than non-caregivers over the follow-up period.

To formally test the relationship suggested by the descriptive analysis, I estimate a Cox PH model of divorce risk. The results are presented in Table 2. This table includes estimation sample to person-waves within each individual's risk set, i.e., observations prior to the divorce event (or up to censoring) which is consistent with standard survival-analysis convention, which therefore reduces the sample shown in the table. Column (1) shows the baseline model without controls. The coefficient on Caring for Parents is 0.419 and significant at the 1% level. This estimate corresponds to an HR of 1.521. Thus, caregivers face a 52.1% higher hazard of divorce than non-caregivers. Columns (2)–(4) sequentially add individual, household, and parental characteristics. The coefficient remains stable, ranging from 0.408 to 0.423. The corresponding HRs range from 1.504 to 1.526. The estimates remain statistically significant at the 1% level. Overall, the results suggest that caregiving for aging parents is associated with approximately 51% higher hazard of divorce .

5.2 Robustness Check

In this section, I conduct several robustness checks to assess the reliability of the main results.

5.2.1 Alternative matching methods

First, I re-estimate the main specification using two alternative matching methods: coarsened exact matching (CEM) and Mahalanobis distance matching. These two methods differ from mai PSM in how they define similarity between caregivers and non-caregivers.

PSM reduces all covariates into a single scalar, the propensity score, and matches individuals based on the closeness of this score. CEM instead matches individuals directly on coarsened bins of the covariates themselves, without reducing them to a single score, and retains any unmatched strata rather than dropping them entirely from the sample. Mahalanobis matching also avoids collapsing covariates into a single score; instead, it matches individuals based on the multivariate Mahalanobis distance across the full covariate space, which accounts for the correlation structure among covariates.

The result is reported in Table 3 with column 1 is result from main PSM, column 2 and 3 are CEM and Mahalanobis results, respectively. Under CEM, the coefficient on caregiving for parents is 0.304 (HR = 1.356), statistically significant at the 5% level. This is smaller in magnitude than the PSM estimate, likely reflecting the larger, less tightly matched sample retained under CEM (35,544 observations). Under Mahalanobis distance matching, the coefficient is 0.414 (HR = 1.513), virtually identical to the PSM estimate. Across both alternative methods, the coefficient remains positive and statistically significant, confirming that caregiving for aging parents is associated with an increased hazard of divorce regardless of the matching approach used.

5.2.2 The strength of unobservable covariates

Second, to assess the sensitivity of the main results to unobservable confounding, I compute the E-value associated with the preferred Cox proportional hazards estimate. The E-value was computed following VanderWeele and Ding (2017). The E-value represents the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need to have with both caregiving for parents and divorce, above and beyond the observed covariates, in order to fully explain away the estimated effect. The detailed calculation is presented in Appendix section 6.2

As reported in Table 4, the E-value for the point estimate is 2.396, and the E-value for the lower bound of the 95% confidence interval is 1.583. This implies that an unmeasured

confounder would need to be associated with both caregiving status and divorce by a risk ratio of at least 2.396 to nullify the observed association. This relatively strong requirement suggests that the main finding is less likely to be entirely driven by unobserved confounding.

To put the E-value of 2.396 into context, we compare it with the strength of several potential confounders that we can observe in the HRS. We consider net worth and full-time employment as measures of economic factors; depressive symptoms because they are part of our proposed mechanism and are also a known predictor of divorce; age because it was the most imbalanced covariate before matching; and education because prior research links it to both caregiver selection and divorce risk. For each covariate, we estimate how strongly it is associated with caregiving and how strongly it is associated with divorce. We then combine these two associations to calculate a bounding factor following VanderWeele and Ding (2017).

Appendix Table A5 shows the associations of observed covariates with caregiving and divorce, and presents E-value benchmark of 2.396 from our preferred model. The observed covariates generally show substantially weaker associations with caregiving and, in most cases, with divorce. Although having negative net worth is relatively strongly associated with divorce (HR = 2.56), its association with caregiving is very small (RR = 1.03). Thus, its observed pattern does not resemble the level of association with both exposure and outcome required by the E-value benchmark. Similarly, depressive symptoms, which theory and prior research identify as an important potential confounder, are only modestly associated with caregiving (RR = 1.06) and divorce (HR = 1.37). Education and employment measures also exhibit relatively small associations with caregiving and divorce. Taken together, these comparisons suggest that an unmeasured confounder would need to be substantially more strongly associated with both caregiving and divorce than the observed socioeconomic, psychological, and demographic factors examined here to fully account for the estimated association. This analysis does not rule out confounding by fac-

tors that are not captured in the HRS. Rather, it indicates that the estimated association is unlikely to be explained by an unmeasured confounder with a strength comparable to the commonly observed economic, psychological, or demographic characteristics included in this benchmarking results.

5.2.3 Alternative survival model

I also re-estimate the relationship between eldercare and caregivers' divorce using a Weibull AFT model. It models the log of survival time directly, rather than the hazard rate. The model is specified as follows:

$$\ln(T_i) = \alpha + \beta_1 \, Caregiving_i(t) + \mathbf{X}_i(t)'\boldsymbol{\gamma} + \sigma\epsilon_i \tag{2}$$

where T_i is the time to divorce for individual i, $Caregiving_i(t)$ is the time-varying caregiving indicator, $\mathbf{X}_i(t)$ is the same vector of individual, household, and parental characteristics used in the main specification, and ϵ_i follows a standard extreme value distribution, which implies that T_i follows a Weibull distribution.

The Weibull AFT model assumes that survival times follow a Weibull distribution. This implies a hazard function of the form $h(t) = p\lambda t^{p-1}$, where the hazard is monotonically increasing or decreasing over time depending on the shape parameter p. This differs from the Cox model, which does not impose a parametric form on the baseline hazard $h_0(t)$ and instead assumes only that the hazard ratio between caregivers and non-caregivers remains constant over time.

Because the Weibull AFT model relies on this different functional form assumption, it serves as a useful test of whether the estimated relationship between caregiving and divorce is sensitive to model specification. A consistent result across both models would suggest that the relationship is not an artifact of the semi-parametric Cox framework.

The coefficient of interest, β_1, is interpreted differently than in the Cox model. A negative value of β_1 indicates that caregiving accelerates the time to divorce, meaning care-

givers experience divorce sooner than non-caregivers. The time ratio, $TR = \exp(\beta_1)$, provides a more directly interpretable measure of this association: a TR below one indicates a shorter expected time to divorce associated with caregiving, while a TR above one would indicate a longer expected time to divorce. Standard errors are clustered at the individual level, consistent with the main specification.

Results from Weibull AFT models are reported in Table 5 with no covariates, individual, household, and parental characteristics are sequentially introduced in columns (1) through (4). Across all four specifications, the coefficient on caregiving for parents is negative and statistically significant at the 1% level, ranging from -0.240 to -0.218. Negative coefficient indicates that caregiving accelerates the time to divorce. That is, caregivers experience divorce sooner than non-caregivers. The corresponding time ratios (TR), calculated as $TR = \exp(\beta)$, range from 0.787 to 0.804, implying that caregiving is associated with a 19.6% to 21.3% reduction in the expected time to divorce relative to non-caregivers, holding covariates constant.

The Weibull AFT results support the main Cox PH findings in Table 2. Both specifications point to the same underlying relationship: caregiving for aging parents is associated with an earlier onset of divorce. The consistency of the results strengthens confidence that the observed association is not an artifact of a particular survival model specification.

5.3 Mechanism

5.3.1 Conceptual Framework

The key finding supports a clear conclusion: caring for aging parents is associated with a higher hazard of divorce among caregivers. To understand the mechanism behind this result, I turn to a conceptual framework grounded in Becker (1974). The central idea is that individuals remain married when the surplus from staying married exceeds the value of the best available alternative, such as separation or divorce. This surplus de-

pends on several inputs that caregiving plausibly disrupts such as time available for the spousal relationship, mental health, and employment. Time constraints and psychological strain can erode how spouses evaluate the quality of their relationship, while reduced employment attachment can lower household resources, both of which shrink the marital surplus and increase the likelihood of divorce. Appendix 6.4 presents the formal model underlying this framework.

The following section presents results on marital quality evaluation, mental health, and labor force participation as potential mechanisms linking caregiving to divorce.

5.3.2 Marital Quality Evaluation

I first examine whether deteriorating marital quality operates as a mechanism linking caregiving to divorce through four measures of spousal evaluation. These measures capture how understanding, open, close a caregiver perceives their spouse to be and how often the spouse lets them down. Each measure is coded on a four-point scale, ranging from "A Lot" to "Not at All." From each ordered logit model, I predict the marginal effect of caregiving on the probability of reporting the highest outcome category, "A Lot." Table 6 reports these marginal effects.

The estimated coefficients are directionally consistent with the hypothesis that caregiving erodes marital quality. The largest decline is for perceived closeness, where caregiving lowers the predicted probability of reporting "a lot" by 3.86 percentage points, significant at the 5% level. This is followed by a 3.13 percentage point decline for understanding, significant at the 10% level. The decline for openness is smaller, at 1.65 percentage points, and is not statistically significant. Caregiving is also associated with a 0.49 percentage point higher predicted probability of feeling "a lot" let down by the spouse, significant at the 10% level, though this effect is small in magnitude. All four coefficients therefore point in the direction expected if caregiving strains the marital relationship, and three of the four are statistically significant at conventional levels.

Statistical power remains a concern. The observed power for these estimates ranges from 0.160 to 0.623, below the conventional 0.80 threshold for every measure. The imprecise estimate for openness likely reflects limited power rather than a true absence of effect: its observed power is only 0.160, and the sample size required to detect an effect of this magnitude at 80% power is 46,560 observations, about 8.6 times the 5,425 observations available. For the other three measures, the required sample sizes range from 8,243 to 12,116 observations, roughly 1.5 to 2.2 times the 5,404 to 5,425 observations available.

These results provide suggestive evidence that declining spousal evaluation, especially perceived closeness and understanding, is one channel through which caregiving may increase the risk of divorce. Because the significant estimates are only moderately precise and the analysis is underpowered, this mechanism should be interpreted with caution and requires further investigation in larger samples.

5.3.3 Caregivers' mental health and labor participation

I examine two additional channels through which caregiving may affect marital outcomes: mental health and labor market attachment. These outcomes are motivated by the literature on informal caregiving, which documents substantial psychological and financial strain among caregivers. Poorer mental health may weaken marital relationships by reducing caregivers' emotional engagement with their spouses. Reduced labor market attachment may increase financial strain within the household. Both mental health and financial strain are important predictors of marital instability.

I estimate OLS regressions of each outcome on caregiving status, controlling for the same set of covariates used in the main analysis. Table 7 reports the results. The results strongly support the mental-health channel, but provide little evidence for the employment channel.

Caregiving is associated with a 0.128-point increase in the CES-D depressive symptoms score, statistically significant at the 1% level. Relative to the sample mean of 1.219,

this corresponds to an increase of roughly 10.5% in depressive symptoms. This estimate is well powered, with an observed power of 0.964, providing strong evidence that caregiving meaningfully worsens caregivers' mental health.

By contrast, the estimates for employment are small and statistically insignificant. Caring for parents is associated with a 1.30 percentage-point decrease in the probability of working full time, although the estimate is not statistically significant. Similarly, caregiving is associated with a 0.44 percentage-point increase in the probability of not working, but this estimate is also statistically insignificant. Thus, both estimates point toward slightly weaker labor-market attachment among caregivers, but the evidence is insufficient to conclude that caregiving meaningfully affects employment. One possible explanation is that these null results reflect limited statistical power, rather than a true absence of effect. Observed power is only 0.321 for full-time employment and 0.082 for non-employment, and the required sample sizes to detect effects of this magnitude at conventional power (68,148 and 559,039 observations, respectively) far exceed the 19,377 observations available.

Collectively, these results point to deteriorating mental health as a robust and well-powered mechanism linking caregiving to marital strain and, ultimately, divorce. The evidence on employment effects remains inconclusive, as the analysis is underpowered to detect effects of a plausible magnitude in either direction.

5.4 Heterogeneity

The effect of caregiving on marital outcomes may vary across individuals due to differences in household financial resources, time constraints and coping capacity. To investigate this heterogeneity, I study whether the impact of caregiving differs by household net worth level, caregivers' gender, education level, race, employment status and household structure. I estimate heterogeneous effects by estimating the effect of caregiving indicator for each subgroup separately. The results are visually presented in Figure A4 and in

more detailed in Table A4. Overall, the results indicate that the effect of caregiving is positive and statistically significant across all groups, but its magnitude varies depending on individuals' characteristics.

First, examining heterogeneity by wealth, caregiving is associated with a significant increase in divorce hazard among individuals with positive net worth (HR = 1.586, significant at the 1% level), whereas the effect among those without positive net worth (HR = 1.311) is smaller and not statistically significant, though this subgroup is also considerably smaller.

Second, examining gender differences, caregiving is associated with a substantially larger increase in the hazard of divorce among men than among women. The estimated hazard ratio is 1.942 for male caregivers, compared to 1.262 for female caregivers, and only the male estimate is statistically significant (at the 1% level). This suggests that men may be less able to absorb the stress associated with caregiving responsibilities for a parent within the marital relationship, potentially reflecting a larger deviation from expected gender roles when men take on caregiving duties.

Third, heterogeneity by education reveals that the effect is significant across both education groups, but somewhat larger among caregivers with a high school degree or higher, where the hazard ratio is 2.180 (significant at the 10% level), compared to 1.505 among those with less than a high school education (significant at the 1% level). The wider confidence interval for the higher-education group reflects its smaller subgroup size.

Fourth, the results reveal meaningful heterogeneity by race. The estimated hazard ratio is 1.552 among White caregivers and 2.005 among caregivers of other races, both statistically significant, while the effect among Black caregivers (HR = 1.133) is smaller and not statistically significant. This suggests that the impact of caregiving on marital instability may not operate uniformly across racial groups in this sample, though the imprecision of the Black-caregiver estimate limits firm conclusions.

Fifth, differences by family structure are also notable. Caregivers with more than one

child exhibit a statistically significant increase in divorce hazard (HR = 1.484, significant at the 1% level), while caregivers with only one child show a larger point estimate (HR = 1.783) that is not statistically significant, likely due to the small size of this subgroup.

Finally, turning to employment status, the effect of caregiving is similar in magnitude between those working full time (HR = 1.483, significant at the 10% level) and those not working full time (HR = 1.540, significant at the 5% level). This suggests that caregiving exerts a comparable strain on marital stability regardless of baseline labor market attachment.

In sum, these results suggest that while caregiving increases the risk of divorce on average, the effect is not uniform across the population. The effect appears most robust among male caregivers and caregivers with higher education, while remaining similar across most wealth, race, and employment subgroups. Several subgroup estimates, however, are imprecisely estimated due to limited subgroup sample sizes, and this heterogeneity pattern should be interpreted with that caveat in mind.

6 Conclusion

Using longitudinal data, matching methods, and survival analysis, this paper provides new evidence on whether caring for aging parents is associated with a higher risk of marital dissolution. The findings point to a straightforward conclusion: providing care for aging parents is associated with a higher hazard of divorce. These results are robust across matching approaches, survival specification and strength of unmeasured confounder.

The mechanism analysis points most clearly to caregiver mental health as the channel through which caregiving translates into marital instability, consistent with the broader caregiving stress-process literature (Amirkhanyan and Wolf, 2006; Bookwala, 2009). Evidence on marital quality and employment outcomes is directionally consistent but does not reach statistical significance, likely reflecting limited statistical power rather than the

absence of a true effect. Overall, these results suggest that the marital cost of caregiving operates more directly through psychological strain than through observable changes in relationship quality or labor supply.

The association between caregiving and divorce hazard is not homogeneous across the caregiver population. The increase in divorce hazard is statistically significant and larger among male caregivers and individuals with higher educational attainment. This heterogeneity can be understood through two complementary lenses.

The first lens is suggested with the conceptual framework developed in Appendix 6.4. Because divorce occurs when the marital surplus falls below the value of the best outside alternative, the same caregiving shock should translate into a larger increase in divorce risk for individuals who already have a higher-valued outside option. Men plausibly fall into this category: men typically face more favorable remarriage markets. For this group, a smaller erosion of marital surplus is needed to cross the threshold into divorce, which is consistent with the larger estimated hazard ratios observed for male and highly educated caregivers.

The second lens is that caregiving disrupts the social or economic role these groups are expected to occupy, over and above its effect on the general caregiver population. For male caregivers, the strain may stem from two related sources. First, stepping into a role that society has traditionally expected women to fill runs against long-standing gender expectations, and this role reversal alone can generate stress (Calasanti, 2010). Second, the way men tend to handle caregiving can create distance from their spouse: rather than treating caregiving as an experience to share, men often approach it as a problem to solve (Calasanti and King, 2007). As their focus becomes consumed by the ”task” of caregiving, they may unintentionally create emotional distance from their spouse - much like someone who becomes so absorbed in work that they neglect their family.

These findings carry several implications for family caregiving policy. Divorce has traditionally been absent from the calculus of caregiving costs, which has focused on care-

givers' physical health, mental health, and labor market outcomes. This study's findings suggest that policymakers should treat family stability as an additional, previously overlooked cost of informal caregiving when designing long-term care policy. For example, current U.S. long-term care policy has deliberately shifted spending away from nursing homes and toward home-based care, known as Medicaid "rebalancing" (Sharille, 2026). While framed as cost-saving, this shift also pushes more caregiving responsibility onto family members (Thunell et al., 2025). If this greater reliance on family caregiving also raises the risk of divorce, as this study suggests, then these cost-saving policies may carry a hidden, offsetting cost: family instability. Policymakers should therefore treat family stability as an additional cost of informal caregiving when evaluating the true efficiency of aging-in-place policy.

Because mental health deterioration emerges as the most robust mechanism linking caregiving to marital dissolution, expanding access to caregiver mental health support should be a policy priority. Programs such as respite care and caregiver counseling already exist under the National Family Caregiver Support Program, which provided counseling and training services to over 100,000 caregivers and respite care to more than 604,000 caregivers nationwide (National Family Caregiver Support Program, 2026). However, this reach remains modest relative to the estimated 63 million family caregivers in the United States (for Caregiving, n.d.). This means the program currently serves roughly 1 percent of family caregivers with counseling services. Expanding the scale of these mental health resources may improve not only caregivers' individual wellbeing but also the stability of their marriages, strengthening the case for sustained investment in support specifically tailored to family caregivers.

The pronounced effect among male caregivers suggests that caregiver support services should be adapted to better serve these caregivers. Male caregivers remain largely unrecognized in caregiving research and services, and are more reluctant than women to seek help from family, friends, or community healthcare services (Akouamba et al.,

2025). This reluctance to seek support may partly explain why existing caregiver support services produce different outcomes for male versus female caregivers (Chen, 2014). Outreach efforts, peer-support groups, and program materials should be tailored to men's more pragmatic, task-oriented approach to caregiving. For example, framing paid or formal assistance as a logical resource to delegate specific tasks rather than as an admission of inability to cope.

The elevated risk among caregivers with higher education level who have a higher opportunity of reduced labor force attachment points to a need for policies that ease the work-care tradeoff. Paid family leave for caregiving responsibilities, rather than only for childbirth, would reduce the financial trade-off caregivers currently face under unpaid leave provisions such as the Family and Medical Leave Act. Similarly, workplace flexibility policies, such as remote work or flexible scheduling, could help caregivers maintain labor force attachment without fully sacrificing their income or professional identity.

As population aging continues to increase the demand for informal elder care, these implications might grow more pressing. The rising incidence of gray divorce documented in recent decades may be partly connected to the growing caregiving burden facing today's middle-aged and older adults. Addressing the family-level costs of caregiving identified in this study is therefore not only a matter of supporting individual caregivers, but of safeguarding family stability as the demand for long-term care continues to rise.

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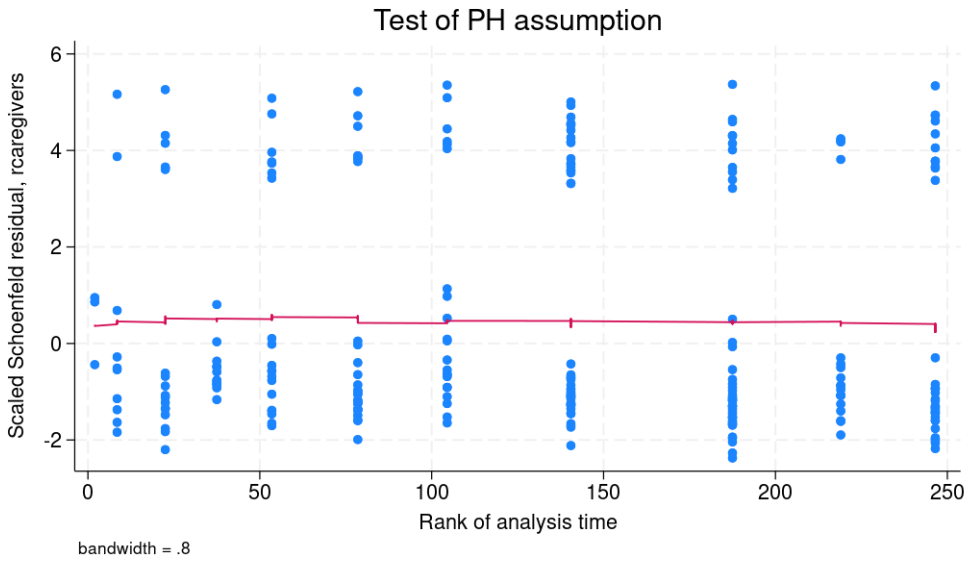
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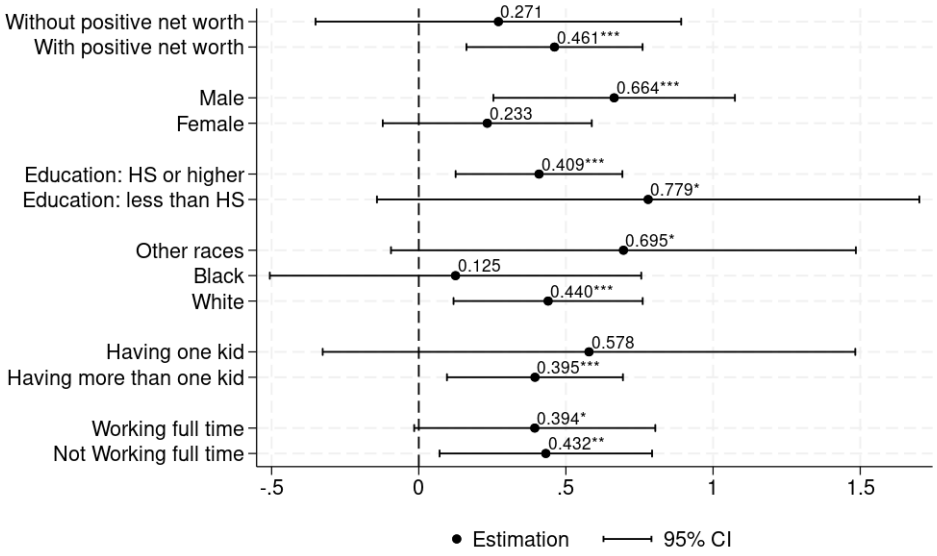
Figures

Figure 1: Test of the Proportional Hazards Assumption



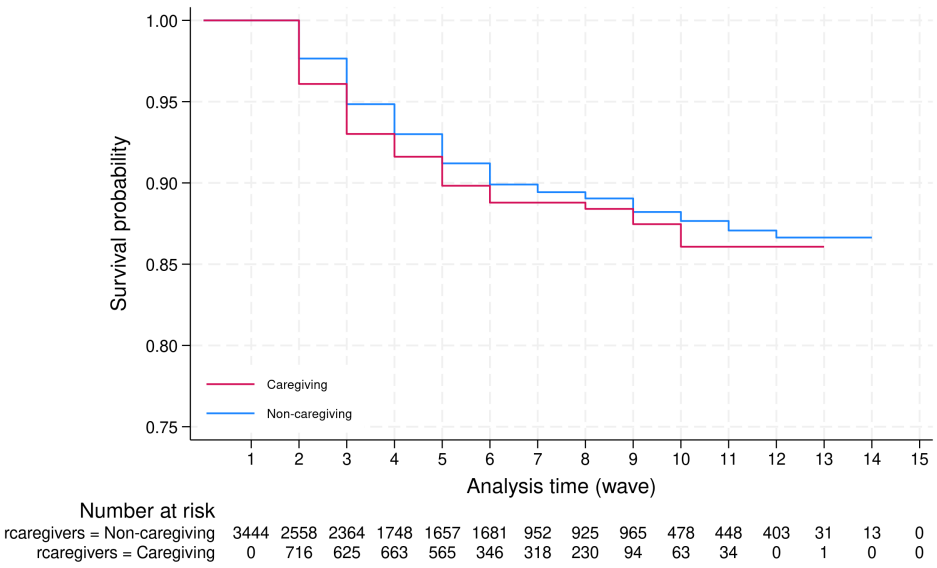
Notes: The figure plots scaled Schoenfeld residuals for the caregiving-for-parents indicator against the rank of analysis time. The solid line represents a smoothed estimate of the residuals. A relatively flat smoothed curve provides no graphical evidence of systematic time variation in the effect of caregiving for parents, consistent with the proportional hazards assumption.

Figure 2: The effect of caring for parents on divorce



Notes: The figure presents subgroup-specific estimates of the effect of caring for parents on the hazard of divorce. Each point represents the estimated coefficient for the corresponding subgroup, with horizontal bars showing the 95% confidence intervals.

Figure 3: Kaplan-Meier Survival Curves for Divorce by Caregiving Status



Notes: The figure presents Kaplan-Meier survival curves for divorce by caregiving status. The survival probability represents the probability of remaining married over analysis time. The curves compare individuals who care for their parents with those who do not. A lower survival probability indicates a higher cumulative incidence of divorce.

Tables

Table 1: Summary Statistics

	All sample		Caregivers		Non-caregivers			
	Mean	(SD)	Mean	(SD)	Mean	(SD)	p-val	Abs. Mean Diff.
Outcome								
Divorce	0.044	(0.206)	0.040	(0.197)	0.050	(0.218)	0.001	0.010
Individual characteristics								
Depressive symptoms score (CES-D)	1.226	(1.852)	1.229	(1.853)	1.222	(1.849)	0.777	0.007
Age (years)	58.524	(8.518)	58.960	(8.270)	57.923	(8.812)	0.000	1.037
Female	0.631	(0.482)	0.641	(0.480)	0.618	(0.486)	0.001	0.023
White	0.786	(0.410)	0.791	(0.407)	0.779	(0.415)	0.044	0.011
Black	0.118	(0.322)	0.116	(0.320)	0.120	(0.325)	0.372	0.004
Other races	0.096	(0.295)	0.093	(0.291)	0.101	(0.301)	0.068	0.007
Foreign-born	0.126	(0.332)	0.120	(0.325)	0.134	(0.340)	0.003	0.013
Less than high school	0.088	(0.283)	0.089	(0.285)	0.086	(0.280)	0.420	0.003
High school degree	0.040	(0.195)	0.036	(0.187)	0.044	(0.205)	0.005	0.008
At least some college	0.626	(0.484)	0.635	(0.481)	0.615	(0.487)	0.002	0.020
Full-time employment	0.461	(0.498)	0.445	(0.497)	0.483	(0.500)	0.000	0.038
Parental characteristics								
Father's education (years)	10.611	(4.093)	10.661	(4.073)	10.542	(4.120)	0.034	0.119
Mother's education (years)	10.860	(3.670)	10.924	(3.612)	10.772	(3.747)	0.003	0.152
Household characteristics								
Number of children in household	2.960	(1.893)	2.991	(1.846)	2.917	(1.956)	0.008	0.073
Log net worth	11.425	(3.558)	11.520	(3.527)	11.295	(3.597)	0.000	0.225
Person-by-wave observations	21,767		12,602		9,165			
Individuals	3,444		1,722		1,722			

Notes: This table summarizes all person-wave observations in the matched sample, including waves observed after a divorce event has occurred for a given individual. Each caregiver is matched to one non-caregiver using propensity score matching with nearest-neighbor matching without replacement. The matching procedure is based on individual, parental, and household characteristics. Standard deviations are reported in parentheses. Binary variables are reported as proportions. The p-values correspond to tests of mean differences between caregivers and non-caregivers. Absolute mean differences are also reported.

Table 2: Effect of Caring for Parents on Divorce

	(1)	(2)	(3)	(4)
	Divorce Hazard			
Caring for Parents	0.419***	0.408***	0.423***	0.415***
(SE)	(0.136)	(0.138)	(0.137)	(0.137)
HR	1.521	1.504	1.526	1.514
Individual characteristics		x	x	x
Household characteristics			x	x
Parental characteristics				x
Person by wave obs	17,507	17,233	17,233	17,233

Notes: The table reports estimates from Cox proportional hazards models examining the association between caregiving for parents and the hazard of divorce. The estimation sample is restricted to each individual's risk set: person-waves following a divorce event are excluded, consistent with standard survival-analysis convention. Each column incrementally adds sets of control variables. The analysis uses person-by-wave observations, with analysis time measured in survey waves. The reported coefficient is the log hazard ratio, $\beta = \log(HR)$, where a positive coefficient indicates a higher hazard of divorce associated with caring for parents. The hazard ratio is calculated as $HR = \exp(\beta)$ and is reported below the coefficient; an HR greater than 1 indicates an increased hazard of divorce. Standard errors are clustered at the individual level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Robustness Check: Alternative Matching Methods

Matching Method	PSM	CEM	Mahalanobis
Dependent variable	Divorce hazard		
Caregiving for parents	0.415***	0.304**	0.414***
(SE)	(0.137)	(0.122)	(0.139)
HR	1.514	1.356	1.513
Individual characteristics	x	x	x
Household characteristics	x	x	x
Parental characteristics	x	x	x
Person $\times$ wave observations	17,233	35,544	17,395

Notes: The table reports estimates of the association between caregiving for parents and the hazard of divorce across alternative matching methods. All specifications control for individual, household, and parental characteristics. PSM denotes propensity score matching (preferred method), CEM denotes coarsened exact matching, and Mahalanobis Match denotes one-to-one Mahalanobis distance matching. Standard errors are reported in parentheses. The reported coefficients are log hazard ratios, and hazard ratios (HR) are reported below the coefficients. The mean of the dependent variable is the mean divorce hazard in the corresponding matched sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Robustness Check - E-value Sensitivity Analysis

	Cox PH	E-value Analysis	
Caregiving for Parents	0.415***		
(SE)	(0.137)		
Hazard Ratio	1.514	E-value	2.396
95% CI	[1.157, 1.981]	E-value for 95% CI	1.583

Notes: The table reports estimates from the preferred Cox proportional hazards model result and the corresponding E-value sensitivity analysis. The hazard ratio and 95% confidence interval are obtained from the Cox proportional hazards model. The E-value represents the minimum strength of association that an unmeasured confounder would need to have with both caregiving for parents and divorce, conditional on the observed covariates, to explain away the estimated association. The E-value for the 95% confidence interval is calculated using the confidence limit closest to the null. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Robustness Check - Alternative Survival Specification: Weibull AFT Model:

	(1)	(2)	(3)	(4)
		Time to Divorce		
Caregiving for Parents	-0.232***	-0.218***	-0.239***	-0.240***
(SE)	(0.074)	(0.070)	(0.075)	(0.076)
TR	0.793	0.804	0.787	0.787
Individual characteristics		x	x	x
Household characteristics			x	x
Parental characteristics				x
Person × wave observations	17,507	17,233	17,233	17,233

Notes: The table reports estimates from Weibull accelerated failure time (AFT) models as an alternative survival-model specification. The outcome is time to divorce, and the coefficients capture the association between caregiving for parents and the log of survival time. The reported coefficients are log time ratios, and time ratios (TR) are reported below the coefficients. The time ratio is calculated as $TR = \exp(\beta)$; a TR below one indicates a shorter time to divorce associated with caregiving for parents, whereas a TR above one indicates a longer time to divorce. Standard errors are clustered at the individual level and reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Marginal Effects of Caring for Parents on Marital Quality Rating

	A Lot Understanding	A Lot Open	A Lot Close	A Lot Letdown
Caring for Parents	-0.0313*	-0.0165	-0.0386**	0.0049*
(SE)	(0.0167)	(0.0173)	(0.0170)	(0.0026)
Depvar mean	1.68535	1.55690	1.50940	3.31440
Observations	5,425	5,425	5,425	5,404
Observed power	0.466	0.160	0.623	0.470
Required N (Power=0.8, $p < 0.1$)	12,116	46,560	8,243	11,935

Notes: The table reports marginal effects of caring for parents estimated from ordered logit models. The marital quality measures are coded on a four-point scale: 1 = "A Lot," 2 = "Some," 3 = "A Little," and 4 = "Not at All." The marginal effects are evaluated for outcome category 1, corresponding to reporting "A Lot" for each marital relationship perception measure. Each coefficient represents the average change in the predicted probability of reporting "A Lot" associated with caring for parents, holding the other covariates constant. Observed power is the post-hoc power based on the estimated marginal effect and its standard error using a two-sided 5 percent significance level. Required N is the approximate sample size needed to achieve 80 percent power. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Effects of Caring for Parents on Mental Health and Employment

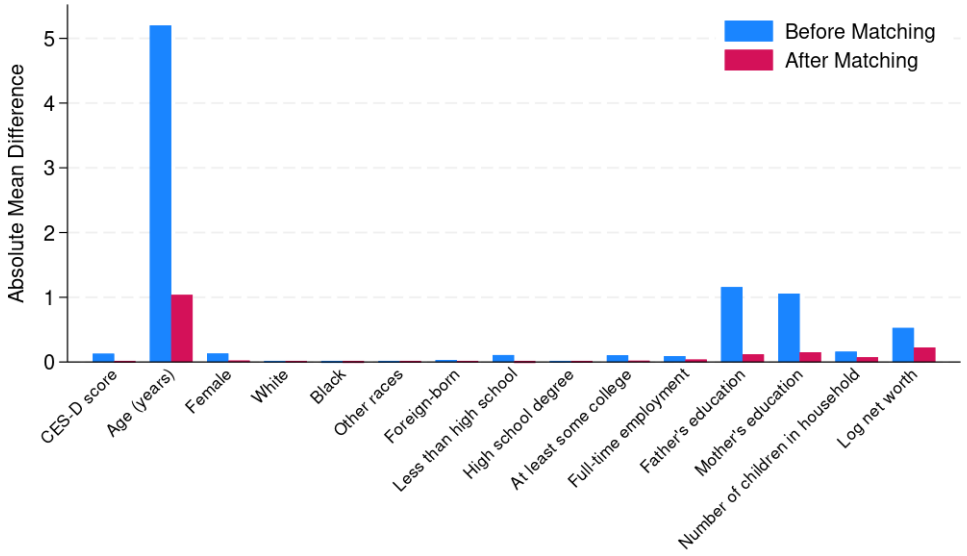
	Mental Health (score)	Working full time	Not working
Caring for Parents	0.1281***	-0.0130	0.0044
(SE)	(0.0341)	(0.0087)	(0.0085)
Person × wave observations	19,020	19,377	19,377
Depvar mean	1.21930	0.47696	0.42932
Observed power	0.964	0.321	0.082
Required N (Power=0.8, $p < 0.1$)	10,591	68,148	559,039

Notes: The table reports estimates from OLS regressions of mental health and employment outcomes on caring for parents. The dependent variable in the first column is the CES-D depressive symptoms score ranges from 1 to 8, with higher scores indicating poorer mental health., while the second and third columns indicate working full time and not working, respectively. Standard errors are reported in parentheses. Observed power is the post-hoc power based on the estimated coefficient and standard error using a two-sided 5 percent significance level. Required N is the approximate sample size needed to achieve 80 percent power based on the observed effect size and standard error.

Appendix A

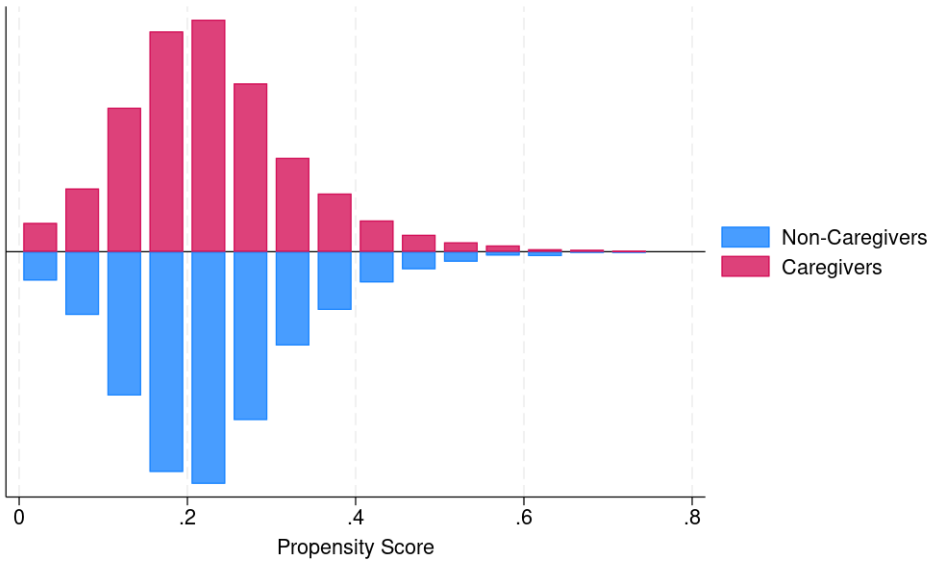
6.1 Figures

Figure A1: Divorce Sample: Covariate Absolute mean difference



Notes: The figure plots covariate absolute Mean difference before and after using PSM to find a comparable non-caregivers group.

Figure A2: Divorce Sample: PSM distribution by caregiving status



Notes: The figure plots propensity score distribution by caregiving group.

6.2 Tables

Table A1: Summary Statistics Before Matching

	All sample		Caregivers		Non-caregivers			
	Mean	(SD)	Mean	(SD)	Mean	(SD)	p-val	Abs. Mean Diff.
Outcome								
Divorce	0.037	(0.189)	0.046	(0.208)	0.035	(0.183)	0.000	0.011
Individual characteristics								
Depressive symptoms score (CES-D)	1.357	(1.890)	1.259	(1.862)	1.388	(1.898)	0.000	0.129
Age (years)	66.151	(11.227)	62.152	(9.546)	67.352	(11.415)	0.000	5.199
Female	0.535	(0.499)	0.637	(0.481)	0.505	(0.500)	0.000	0.132
White	0.810	(0.392)	0.819	(0.385)	0.807	(0.394)	0.000	0.012
Black	0.128	(0.334)	0.124	(0.330)	0.129	(0.335)	0.006	0.005
Other races	0.062	(0.241)	0.056	(0.231)	0.063	(0.244)	0.000	0.007
Foreign-born	0.129	(0.335)	0.105	(0.306)	0.136	(0.343)	0.000	0.032
Less than high school	0.219	(0.414)	0.136	(0.343)	0.244	(0.430)	0.000	0.108
High school degree	0.048	(0.213)	0.040	(0.195)	0.050	(0.218)	0.000	0.010
At least some college	0.427	(0.495)	0.507	(0.500)	0.403	(0.491)	0.000	0.104
Full-time employment	0.271	(0.445)	0.341	(0.474)	0.250	(0.433)	0.000	0.091
Parental characteristics								
Father's education (years)	9.277	(3.891)	10.151	(3.820)	8.994	(3.872)	0.000	1.157
Mother's education (years)	9.527	(3.624)	10.328	(3.494)	9.271	(3.627)	0.000	1.058
Household characteristics								
Number of children in household	3.413	(2.156)	3.289	(1.984)	3.449	(2.204)	0.000	0.161
Log net worth	11.245	(3.370)	11.651	(3.104)	11.124	(3.437)	0.000	0.527
Person-by-wave observations	191,110		44,152		146,958			
Individuals	26,911		4,614		22,297			

Notes: This table reports summary statistics for the sample before matching and separately by caregiving status. Standard deviations are reported in parentheses. Binary variables are reported as proportions. The p-values correspond to tests of mean differences between caregivers and non-caregivers. Absolute mean differences are also reported.

Table A2: Covariate Balance after Propensity Score Matching

	Mean		% Bias	t-test		Var. Ratio
Variable	Treated	Control		t	p-value	V(T)/V(C)
Depressive symptoms (score)	1.258	1.250	0.5	0.14	0.888	0.95
Mother's education (years)	10.949	10.821	3.4	1.00	0.320	0.94
Father's education (years)	10.739	10.652	2.1	0.61	0.539	0.99
Less than high school	0.095	0.089	1.8	0.53	0.596	.
High school degree	0.039	0.045	−2.6	−0.76	0.445	.
Some college or more	0.632	0.617	3.1	0.91	0.360	.
Age (years)	51.528	51.426	1.7	0.50	0.617	0.80*
Number of children	2.850	2.796	3.0	0.87	0.386	0.86*
White	0.762	0.743	4.4	1.30	0.193	.
Female	0.620	0.612	1.8	0.53	0.599	.
Foreign-born	0.142	0.150	−2.5	−0.72	0.469	.
Working full-time	0.610	0.629	−3.8	−1.12	0.261	.
Log household net worth	11.007	10.966	1.1	0.33	0.741	0.98

Notes: This table reports covariate balance after propensity score matching. Bias is measured as the standardized percentage difference in means between treated and control groups. Asterisks (*) indicate variance ratios outside the acceptable range of [0.91, 1.10].

Table A3: Proportional-Hazards Test for Caregiving

Covariate	Correlation (ρ)	χ^2	p-value
Caregiving	-0.02912	0.26	0.6078
Global test		16.22	0.3674

Note: The proportional-hazards assumption is assessed using Schoenfeld residuals with the rank of analysis time as the time function. The correlation (ρ) measures the correlation between the scaled Schoenfeld residuals and the rank of analysis time. The global test evaluates the proportional-hazards assumption jointly across all covariates. A p-value greater than 0.05 indicates no evidence of a violation of the proportional-hazards assumption.

Table A4: Heterogeneous Effects of Caregiving for aging parents on Caregivers' Divorce

Group	Coef	SE	HR	Individuals	Person-wave Obs
Wealth					
Without positive net worth	0.271	(0.317)	1.311	630	1,111
With positive net worth	0.461***	(0.153)	1.586	3,225	16,109
Gender					
Male	0.664***	(0.209)	1.942	1,253	6,259
Female	0.233	(0.181)	1.262	2,037	10,961
Education					
Education: HS or higher	0.409***	(0.145)	1.505	2,992	15,760
Education: less than HS	0.779*	(0.470)	2.180	298	1,460
Race					
Other races	0.695*	(0.403)	2.005	366	1,615
Black	0.125	(0.322)	1.133	436	1,924
White	0.440***	(0.164)	1.552	2,488	13,681
Family Structure					
Having more than one kid	0.395***	(0.153)	1.484	2,838	14,555
Having one kid	0.578	(0.462)	1.783	403	1,667
Employment					
Working full time	0.394*	(0.209)	1.483	2,427	8,379
Not working full time	0.432**	(0.184)	1.540	2,289	8,841

Notes: This table reports heterogeneous effects of caregiving for parents on caregivers' divorce hazard using Cox proportional hazards models. Coefficients represent log hazard ratios, and hazard ratios (HR) are reported for interpretation. Each row corresponds to a separate subgroup analysis. All specifications include individual, parental and household characteristics. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: E-value Benchmarking Against Observed Covariates

Covariate	**Association with Caregiving**	**Association with Divorce**	**E-value**
Having negative net worth	1.03	2.56	2.396
Depressive symptoms	1.06	1.37	2.396
Age	1.10	0.83	2.396
Less than high school	1.02	1.19	2.396
High school degree	0.89	1.97	2.396
At least some college	1.07	1.14	2.396
Full-time work	0.95	1.00	2.396

Notes: The association with caregiving is reported as an adjusted risk ratio (RR), calculated from predicted probabilities obtained from a logistic regression of caregiving status on each benchmark covariate and the remaining observed covariates. For binary covariates, the RR compares the adjusted predicted probability of caregiving when the covariate equals 1 versus 0. For continuous covariates, the RR compares the predicted probability at one standard deviation above the reference value ($Z = 1$ SD) with that at the reference value ($Z = 0$). The association with divorce is reported as the adjusted hazard ratio (HR) for the benchmark covariate from a Cox proportional hazards model that includes caregiving status and the remaining observed covariates. Standard errors are clustered at the individual level. The benchmark covariate is excluded from the control set in each corresponding model. The E-value of 2.396 is the E-value obtained from the preferred model and is used as the benchmark for comparison.

Appendix B

6.3 E-value computation

This section explain E-value computation as a sensitivity analysis framework proposed by VanderWeele and Ding (2017). I compute the E-value associated with the estimated hazard ratio. Intuitively, the E-value asks how strongly an unobserved variable would need to be related to both caregiving status and divorce, over and above the included covariates, before it could account for the entire estimated effect on its own. While this exercise cannot rule out omitted-variable bias outright, it offers a concrete, interpretable benchmark against which the plausibility of such bias can be judged.

The E-value is computed as follows:

$$
\begin{aligned}
E &= \begin{cases} HR + \sqrt{HR(HR-1)}, & HR > 1, \\ \dfrac{1}{HR} + \sqrt{\dfrac{1}{HR}\left(\dfrac{1}{HR} - 1\right)}, & HR < 1, \end{cases} \\
E_{\text{CI}} &= \begin{cases} 1, & HR > 1,\ LL \leq 1, \\ LL + \sqrt{LL(LL-1)}, & HR > 1,\ LL > 1, \\ 1, & HR < 1,\ UL \geq 1, \\ \dfrac{1}{UL} + \sqrt{\dfrac{1}{UL}\left(\dfrac{1}{UL} - 1\right)}, & HR < 1,\ UL < 1. \end{cases}
\end{aligned} \tag{3}
$$

The top expression in Equation 3 gives E, the E-value corresponding to the point estimate of the hazard ratio. A higher value of E means that only a relatively strong hidden confounder could plausibly overturn the estimated association, whereas a low value would signal that even modest confounding could do so.

The bottom expression gives E_{CI}, which applies the same logic to the confidence interval rather than the point estimate alone. To keep the calculation conservative, I use only the confidence bound nearer to the null value of one—the lower bound (LL) when the hazard ratio exceeds one, or the upper bound (UL) when it falls below one. When this bound already spans the null (i.e., the interval includes one), E_{CI} is set to one by construction, since no confounding is needed to render the interval statistically indistinguishable from no effect. Otherwise, E_{CI} is obtained by applying the same E-value formula to whichever bound lies closest to one.

6.4 Conceptual Framework

This section develops a simple conceptual framework to explain why caregiving for aging parents may increase the risk of marital dissolution. The framework builds on the theory of marriage developed by Becker (1974). The basic idea is that individuals remain

married when the benefits of staying married are greater than the benefits of leaving the marriage.

I capture this idea with the concept of marital surplus:

$$S_{it} = U_{it}^{M} - U_{it}^{A}. \tag{4}$$

Here, U_{it}^{M} represents the utility from remaining married, while U_{it}^{A} represents the utility from the best alternative to the current marriage, such as separation or divorce, potentially followed by remarriage. An individual remains married when $S_{it} > 0$, meaning that staying married provides greater utility than the available alternative. Conversely, when $S_{it} < 0$, the alternative to marriage provides greater utility, making marital dissolution more likely.

6.4.1 Utility from remaining married

The benefits of remaining married depend on several aspects of household and individual well-being. In particular, marriage may provide financial resources, time for the relationship, and psychological well-being:

$$U_{it}^{M} = f\big(Y_{it},\, T_{it},\, H_{it}\big), \qquad f_Y, f_T, f_H > 0, \tag{5}$$

where Y_{it} represents household income and other financial resources, T_{it} represents time available for the spousal relationship, and H_{it} represents psychological health. Higher income, more time available for the relationship, and better psychological health are assumed to increase the benefits of remaining married.

6.4.2 How caregiving may reduce marital surplus

Caregiving for an aging parent can create additional financial, time, and emotional demands on the caregiver. These demands may reduce the resources available to the household, reduce time available for the spousal relationship, and increase psychological distress. I represent caregiving as $C_{it} \in \{0, 1\}$ and assume that caregiving can affect these three dimensions as follows:

$$\frac{\partial Y_{it}}{\partial C_{it}} < 0 \qquad \text{(reduced earnings or employment)}, \tag{6}$$

$$\frac{\partial T_{it}}{\partial C_{it}} < 0 \qquad \text{(less time for the spousal relationship)}, \tag{7}$$

$$\frac{\partial H_{it}}{\partial C_{it}} < 0 \qquad \text{(worse psychological health)}. \tag{8}$$

Because each of these changes reduces the benefits of remaining married, caregiving can lower marital utility:

$$\frac{\partial U_{it}^{M}}{\partial C_{it}} < 0. \tag{9}$$

In other words, caregiving can reduce the benefits of staying married by increasing financial pressure, reducing time available for the couple, and increasing emotional or psychological strain. As a result, caregiving can reduce marital surplus and make divorce more likely.

6.4.3 The value of the alternative to marriage

The decision to remain married also depends on how attractive the alternative to marriage is. I represent this outside option as:

$$U_{it}^{A} = g(W_{it}, R_{it}), \qquad g_W, g_R > 0, \tag{10}$$

where W_{it} represents resources that an individual can rely on independently of the marriage, such as personal wealth and earnings capacity, and R_{it} represents prospects in the remarriage market.

Individuals with greater independent financial resources may face lower economic costs from leaving a marriage. Similarly, individuals with greater earning capacity may be better able to support themselves after divorce. Remarriage prospects may also affect the attractiveness of leaving the current marriage. For example, younger individuals and men may, on average, face more favorable remarriage prospects. These factors increase the value of the outside option, U_{it}^{A}.

6.4.4 Caregiving has a larger effect when marital surplus is initially low

The key prediction of the framework is that the effect of caregiving on divorce depends on how close an individual is to the point at which leaving the marriage becomes more attractive than staying.

Before caregiving, consider two individuals who are both married. One has a large marital surplus, meaning that the benefits of remaining married are substantially greater than the benefits of the alternative. The other has a small marital surplus, meaning that the two options provide similar levels of utility. If caregiving reduces the benefits of remaining married by the same amount for both individuals, the second individual is more likely to cross the threshold into negative marital surplus.

Thus, caregiving should have a larger effect on divorce among individuals whose marital surplus is already close to zero before caregiving begins:

$$S_{it} = U_{it}^{M} - U_{it}^{A} \approx 0. \tag{11}$$

The same logic implies that caregiving should have a larger effect among individuals with a higher value of the outside option, U_{it}^{A}. When the outside option is more attractive, the gap between the benefits of staying married and leaving the marriage is smaller. A given reduction in marital utility caused by caregiving is therefore more likely to push the individual toward marital dissolution.

Formally, this prediction can be expressed as:

$$\frac{\partial^2 \Pr(\text{divorce}_{it})}{\partial C_{it}\, \partial U_{it}^{A}} > 0. \tag{12}$$

In substantive terms, this means that the effect of caregiving on divorce risk should be larger for individuals with greater independent wealth, higher earnings capacity, or more favorable remarriage prospects. These characteristics make the alternative to the current marriage more attractive and therefore reduce the amount of caregiving-related strain required to make divorce a more attractive option.